PyTask Pro – Technical Architecture Report

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Domain: Python Developer **Duration:** 15 Days

1. Executive Summary

PyTask Pro is a modular command-line automation suite for extracting structured information from HTML pages, cleaning text payloads with regular expressions, and exporting results to JSON and CSV. The implementation follows the assignment's recommended Python 3.11, OOP, Requests, BeautifulSoup, argparse, file I/O and testing requirements.

2. Architecture

The package is separated into domain models, logging/decorators, scraping, cleaning, exporting and CLI orchestration. This separation keeps responsibilities focused and makes each component independently testable.

```
CLI
 |
 +-- scraper.py ---- Requests + BeautifulSoup
 |       |
 |       +-- Record objects
 |
 +-- cleaner.py ---- regex normalization
 |
 +-- exporter.py --- JSON / CSV
 |
 +-- logger.py ----- timing + logging decorators
 |
 +-- models.py ----- OOP domain layer
```

3. OOP Design

The Record dataclass represents scraped data. BaseProcessor provides an extensible processing abstraction, while RecordProcessor demonstrates inheritance. Encapsulation is used through the protected _name field and a read-only property.

4. Scraping Engine

WebScraper uses a requests.Session, configurable timeouts, retry attempts, rotating user-agent strings and a configurable delay. HTTP 429 responses are handled with a bounded Retry-After wait. Other request failures are retried and ultimately surfaced as a RuntimeError. HTML parsing uses BeautifulSoup selectors.

5. Data Cleaning

TextCleaner normalizes repeated whitespace and removes unsupported control characters using Python's re module. Cleaning is applied consistently to every Record field before export.

6. CLI and File I/O

The argparse interface provides scrape, clean, export and run subcommands. The run command executes the complete scrape → clean → JSON/CSV export pipeline. DataExporter provides JSON round-trip support and CSV generation.

7. Testing

The pytest suite covers regex cleaning, record cleaning, JSON round trips, CSV headers and HTML parsing, including an empty-page case. Tests isolate deterministic parser and file-processing behavior from live network access.

8. Packaging and Installation

pyproject.toml defines the project metadata, Python version, dependencies, optional testing dependencies and the console-script entry point. The package can be installed with pip install -e . inside a virtual environment.

9. Assignment Requirement Mapping

Requirement	Delivered

Core Python / Python 3.11	pyproject.toml and typed Python modules
OOP + inheritance	models.py
Custom decorators	logger.py
Web scraping	scraper.py
HTTP errors / rate limiting	retries, 429 handling, delay
User-agent rotation	ScraperConfig
Regex cleaning	cleaner.py
JSON / CSV	exporter.py
CLI parser	argparse subcommands
Unit testing	pytest tests
Packaging	pyproject.toml
Documentation	README + this report

10. Responsible Use

The scraper is intended for educational use. Operators should only automate sites where access is permitted, follow robots.txt and terms of service, use conservative request rates, and avoid collecting sensitive or restricted information.

11. Conclusion

PyTask Pro provides a complete, maintainable implementation of the internship's core engineering requirements: modular OOP architecture, decorators, resilient web scraping, regex cleaning, CLI automation, JSON/CSV export, testing and packaging.